

Exponential function

Homework

№1. Find an interval on which the maximum value of the function $y = \left(\frac{1}{3}\right)^x$ is equal to 27 and the minimum equals to $\frac{1}{9}$.

№2. Solve an inequality $2^{\frac{1}{x}} > 0$.

№3. Plot the graph of the function:

$$1. \quad y = \left| \frac{1}{2^x} - 1 \right|; \quad 2. \quad y = |2^{-|x|} - 1|; \quad 3. \quad y = \frac{2^{|x|} - 1}{|2^x - 1|}.$$

№4. Determine whether the function $y = (2 + \sqrt{3})^x + (2 - \sqrt{3})^x$ is even or odd.

№5. Determine the range of the function $y = \frac{3^x}{3^x - 9}$.

№6. Given the function $f(x) = 3^x + \frac{27}{3^x}$. What value of a makes $f(x + a)$ an odd function?

№7. What value of a makes the function $f(x) = 4^{-x} + \left(\frac{1}{2}\right)^{x+1} \cdot \frac{5a}{2} + \frac{a^2 + 12}{6}$ be greater than 2 on the interval $[-1; 1]$?